Title: **Practical Data Science with Python COSC2670 Assignment 2**

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I certify that this is all my own original work. If I took any parts from elsewhere, then they were non-essential parts of the assignment, and they are clearly attributed in my submission. I will show I agree to this honor code by typing "Yes": **Yes**

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1. Abstract/Executive Summary

The aim of this report was to segment users based on their interests in various types of points of interest (POIs) using clustering techniques. An exploratory data analysis was conducted, followed by feature scaling, K-Means, and DBSCAN clustering. Overall, the results indicate that K-Means identified 5 optimal clusters with a silhouette score of 0.289. The report concludes that K-Means clustering is recommended for user segmentation based on its higher silhouette score compared to DBSCAN. It is recommended to utilize the K-Means clustering results to tailor marketing strategies and service offerings for different user segments.

2. Introduction

The goal of this analysis is to segment users based on their interests in various types of points of interest (POIs) using clustering techniques. By understanding these segments, we can tailor marketing strategies and service offerings to better meet the diverse needs and preferences of different user groups.

Previous studies have shown that user segmentation based on interests and preferences can lead to more effective targeted marketing and improved customer satisfaction. However, traditional segmentation methods often rely on predefined rules or assumptions, which may not capture the underlying patterns in the data.

Clustering algorithms, such as K-Means and DBSCAN, offer a data-driven approach to user segmentation by identifying groups of users with similar interests based on their activity patterns. These algorithms can uncover hidden structures and relationships within the data, providing valuable insights for businesses to optimize their marketing efforts and product offerings.

3. Methodology

Data Preparation and Cleaning

The dataset 'buddymove_holidayiq.csv' was loaded into a pandas DataFrame. It contains 249 entries and 7 columns: User Id, Sports, Religious, Nature, Theatre, Shopping, and Picnic.

Dataset Overview:

- **User Id**: Unique identifier for each user.
- **Sports**: Number of reviews related to sports attractions.
- **Religious**: Number of reviews related to religious attractions.
- **Nature**: Number of reviews related to nature attractions.
- **Theatre**: Number of reviews related to theatre attractions.
- **Shopping**: Number of reviews related to shopping attractions.
- **Picnic**: Number of reviews related to picnic attractions.

Summary Statistics:

- Sports has the lowest mean (11.99) and the highest standard deviation (6.62).
- Nature has the highest mean (124.52) and the highest range (52 to 318).

Missing Values and Duplicates:

- There are no missing values or duplicate rows in the dataset.

Data Types:

- User Id is of type object, and all other columns are of type int64.

Data Integrity Checks:

- Basic checks confirmed data integrity, ensuring no missing values or duplicates, and appropriate data types were maintained.

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Data Exploration

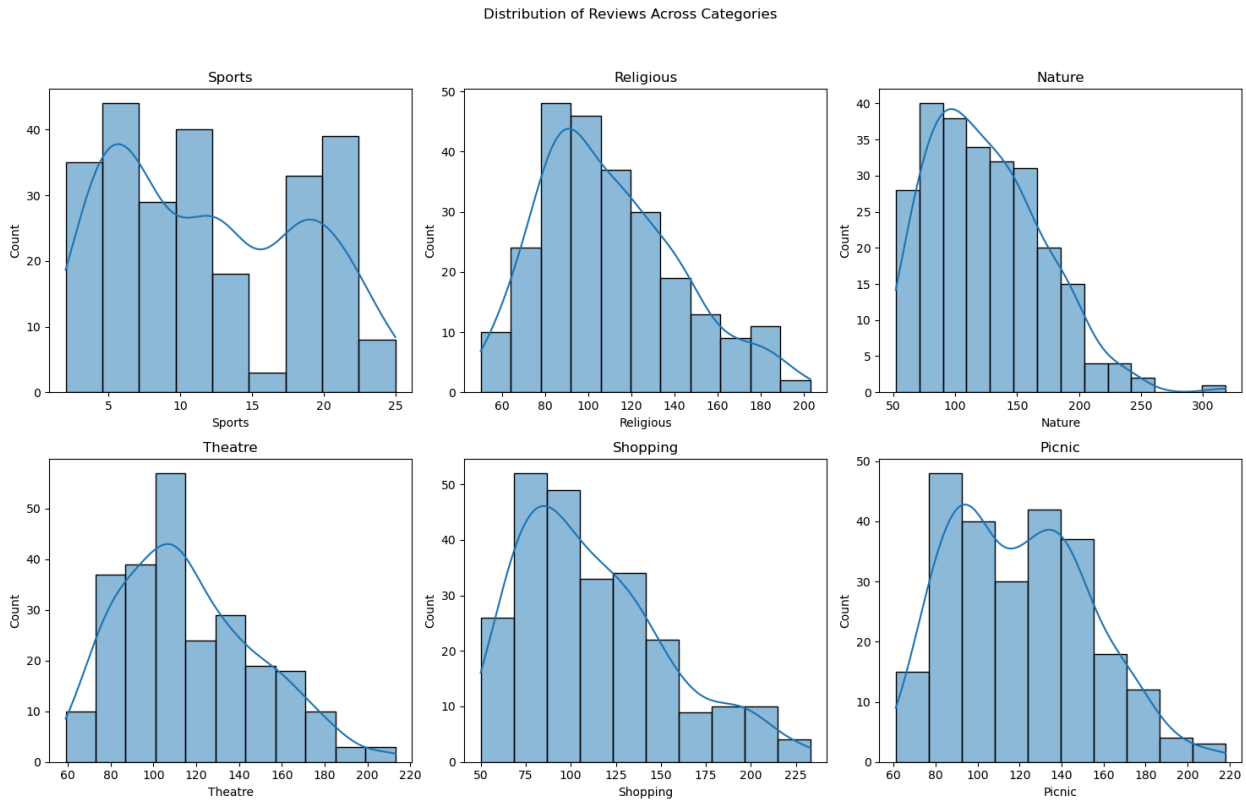
Exploratory data analysis was conducted to gain insights into the dataset:

Graphs and Insights:

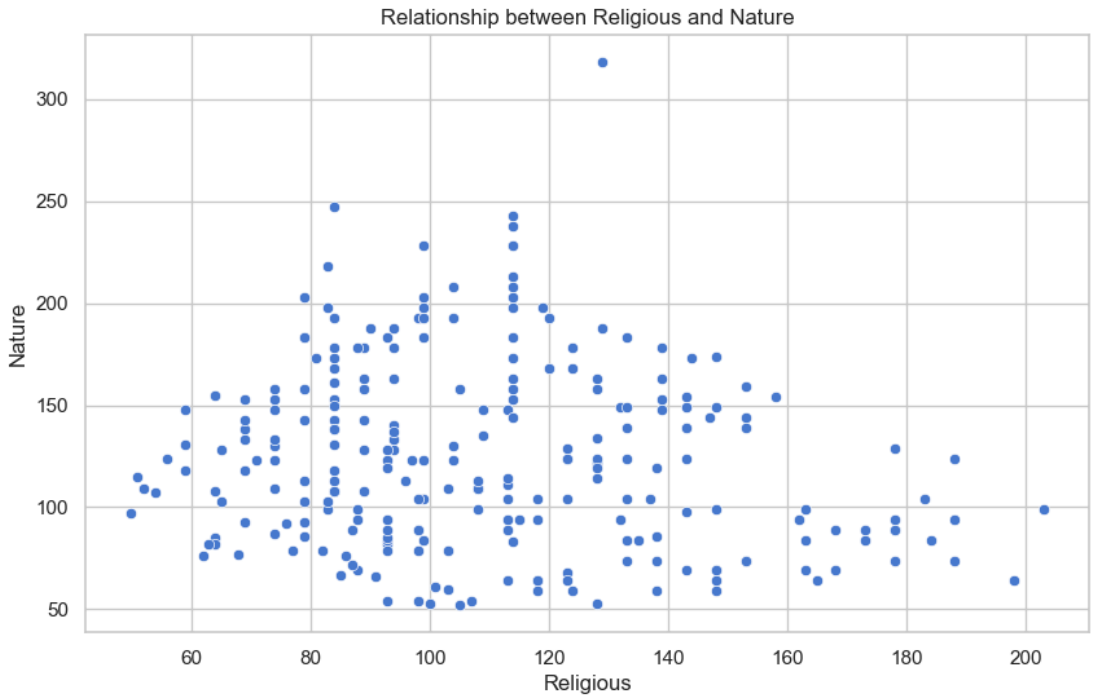
- **Sports Reviews**: Reviews range from 2 to 25, with most users writing 6 to 18 reviews.
- **Religious Reviews**: Reviews range from 50 to 203, concentrated between 84 and 132.
- **Nature Reviews**: Reviews span from 52 to 318, concentrated around 89 to 153.
- **Theatre Reviews**: Reviews range from 59 to 213, mostly between 93 and 138.
- **Shopping Reviews**: Reviews range from 50 to 233, concentrated between 79 and 138.
- **Picnic Reviews**: Reviews span from 61 to 218, mostly between 92 and 143.

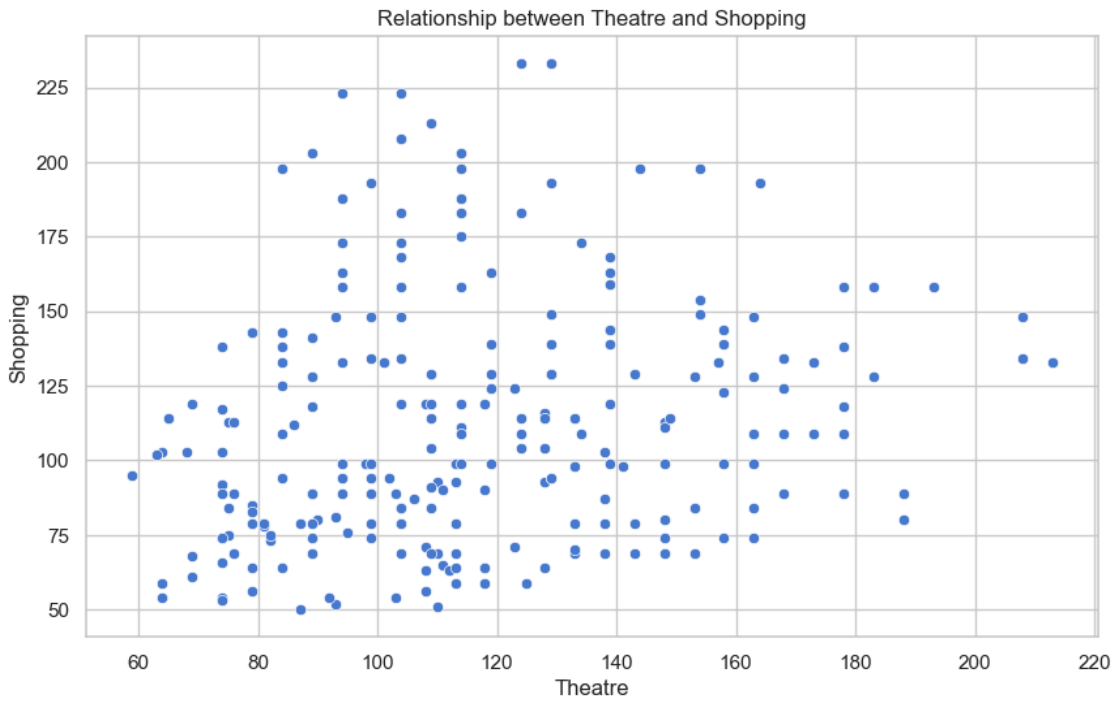
1. **Exploring Each Column**: The distribution of each column (excluding 'User Id') was visualized using histograms with kernel density estimates. The 'Sports' column showed a right-skewed distribution, while 'Religious' and 'Nature' had more spread-out distributions. 'Theatre', 'Shopping', and 'Picnic' exhibited bimodal distributions.

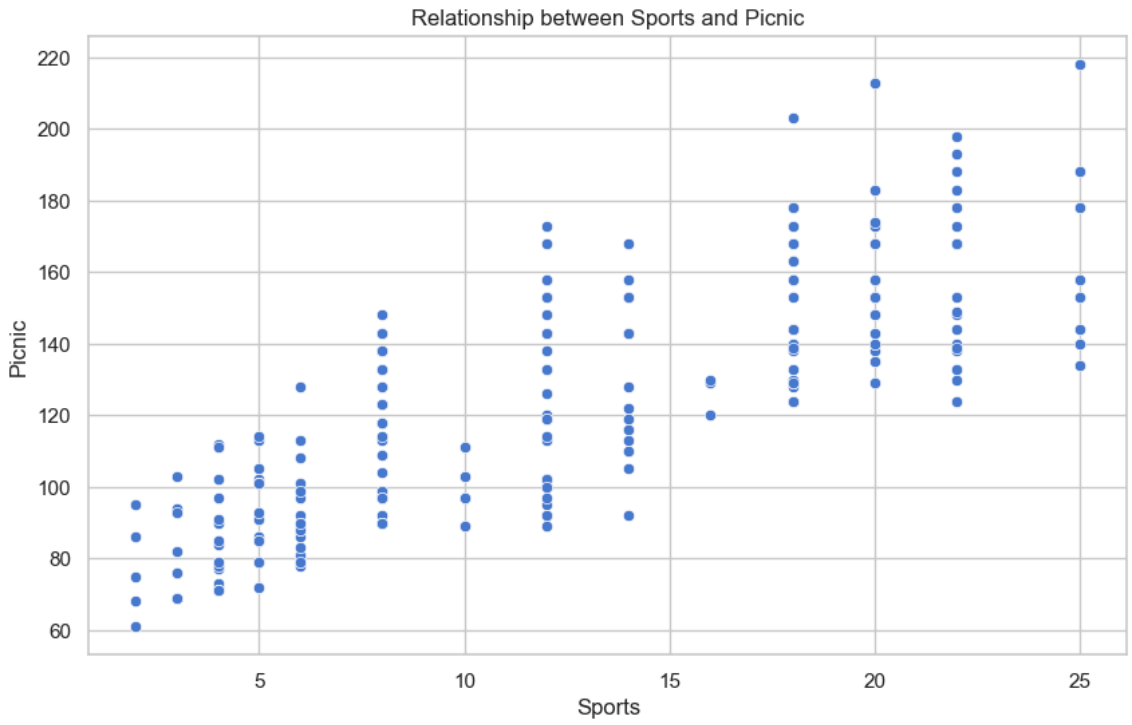
- **Sports**: Most users have low scores, with a few having higher scores.
- **Religious**: Wide distribution with a peak around 100.
- **Nature**: Wide distribution with a peak around 120.
- **Theatre**: Wide distribution with a peak around 110.
- **Shopping**: Wide distribution with a peak around 100.
- **Picnic**: Wide distribution with a peak around 120.



2. **Exploring Relationships Between Pairs of Attributes**: Scatter plots were used to visualize relationships between pairs of attributes. Interesting observations included a weak positive correlation between 'Religious' and 'Nature', a moderate positive correlation between 'Theatre' and 'Shopping', and a weak negative correlation between 'Picnic' and 'Sports'.


Relationship between Religious and Nature
Nature
Religious
300
250
200
150
100
50
60
80
100
120
140
160
180
200


Relationship between Theatre and Shopping
Shopping
Theatre
225
200
175
150
125
100
75
50
60
80
100
120
140
160
180
200
220

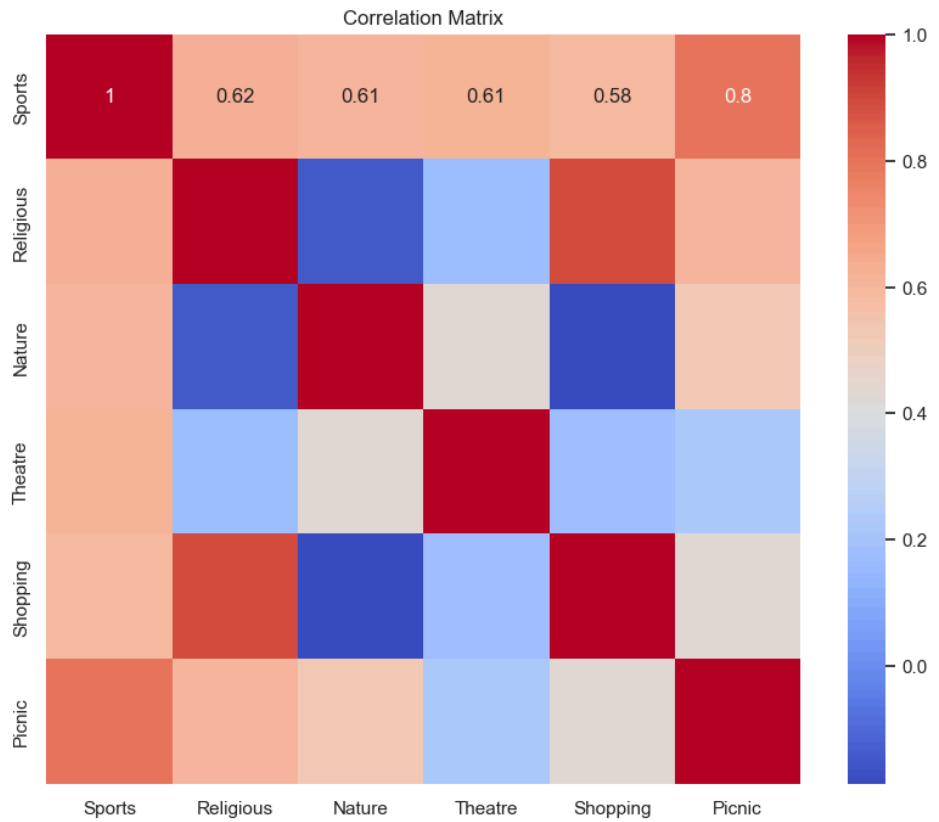


Feature Selection and Scaling

The 'User Id' column was dropped as it is not a relevant feature. The remaining columns were standardized using StandardScaler to ensure equal importance of features during clustering.

Correlation Matrix

A correlation matrix was computed to understand the relationships between features. Moderate positive correlations were observed between 'Religious' and 'Nature', 'Theatre' and 'Shopping', and 'Shopping' and 'Picnic', aligning with the observations from the scatter plots.



Clustering

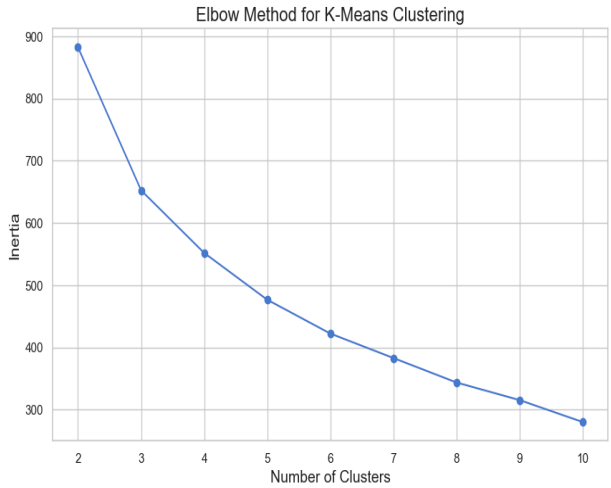
Two clustering algorithms, K-Means and DBSCAN, were employed for user segmentation.

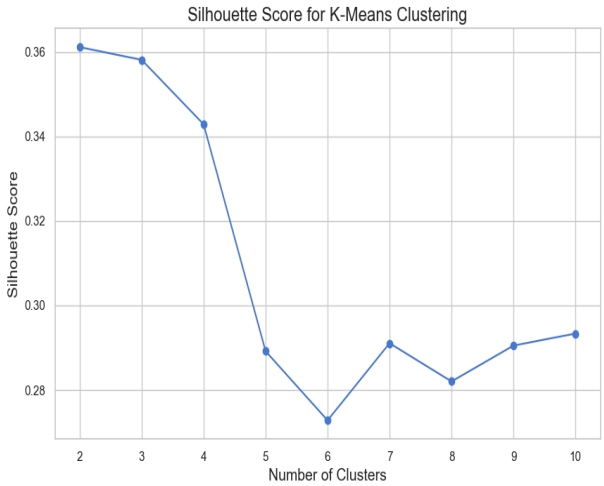
K-Means Clustering

The optimal number of clusters for K-Means clustering was determined using the Elbow method and Silhouette scores. Based on these methods, the optimal number of clusters was chosen as 5.

K-Means clustering was performed with 5 clusters, and the Silhouette score was calculated to evaluate the clustering performance. The Silhouette score for K-Means clustering was 0.289, indicating a reasonable clustering performance.

A confusion matrix was constructed to analyze the meaning of each cluster by looking at the majority of observations in each cluster.

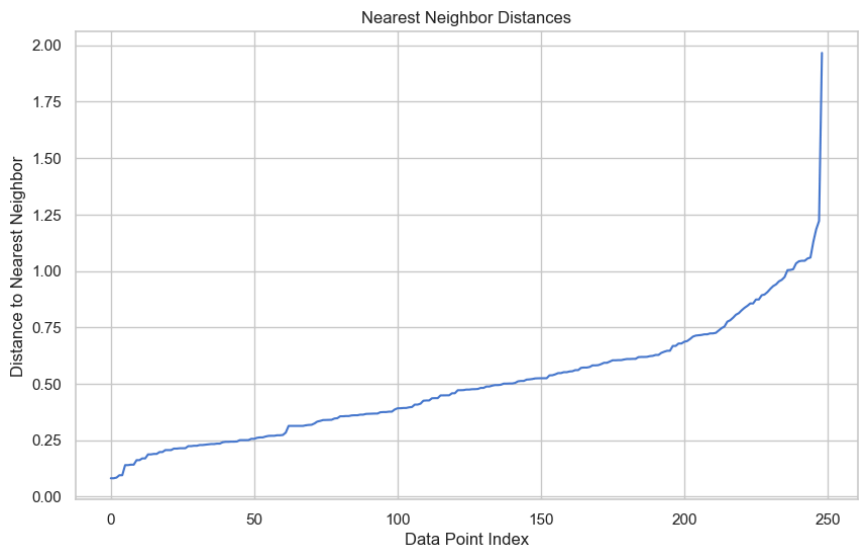




DBSCAN Clustering

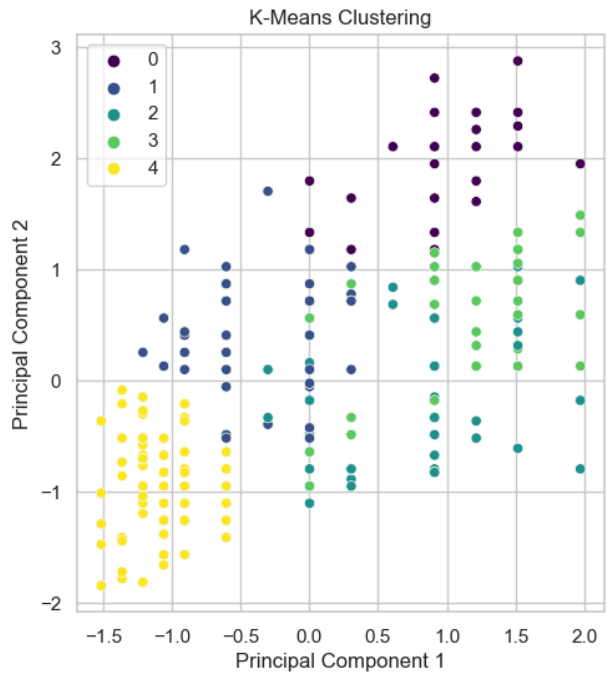
The `eps` parameter for DBSCAN was determined using the k-distance plot, and a suitable value of 0.6 was chosen. The `min_samples` parameter was set to 8 based on domain knowledge and as we might expect user groups with at least 8 members to have similar interests.

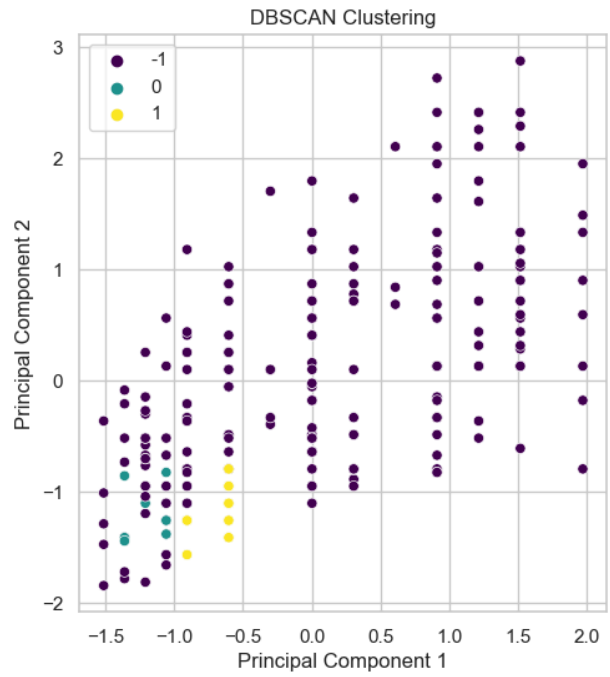
DBSCAN clustering was performed with the chosen parameters, and the Silhouette score was calculated. The Silhouette score for DBSCAN clustering was -0.114, lower than K-Means clustering.



Visualizing Clusters

The clusters identified by both algorithms were visualized using scatter plots, providing a visual representation of the user segments.





4. Results

The key results of the analysis are as follows:

1. **Exploratory Data Analysis**: The distributions of the columns and relationships between pairs of attributes provided insights into the user interests and preferences.

2. **Correlation Matrix**: Moderate positive correlations were observed between certain pairs of attributes, indicating potential relationships between user interests.

3. **K-Means Clustering**: K-Means clustering identified 5 optimal clusters with a Silhouette score of 0.289, indicating a reasonable clustering performance.

4. **DBSCAN Clustering**: DBSCAN clustering had a lower Silhouette score of -0.114 compared to K-Means clustering.

5. **Cluster Visualization**: Scatter plots provided a visual representation of the identified user segments for both clustering algorithms.

5. Discussion

The analysis aimed to segment users based on their interests in various types of POIs using clustering techniques. The exploratory data analysis revealed interesting patterns and relationships within the data, such as the bimodal distributions of certain attributes and the correlations between pairs of attributes.

K-Means clustering emerged as the preferred method for user segmentation, as it achieved a higher Silhouette score compared to DBSCAN. The identified 5 clusters represent distinct user segments with similar interests and preferences. By analyzing the characteristics of each cluster, businesses can gain valuable insights into the diverse needs and preferences of their user base.

The confusion matrix constructed for K-Means clustering provides a deeper understanding of the composition of each cluster, allowing for targeted marketing strategies and tailored service offerings to be developed for each user segment.

Segmentation Benefits:
User segmentation offers several benefits:
Targeted Marketing: Tailor strategies to specific segments for higher engagement.
Product Customization: Develop features or packages for different segments.
Optimized Resource Allocation: Allocate resources efficiently based on segment value.
Improved Customer Retention: Deliver personalized experiences to reduce churn.
Market Expansion Opportunities: Identify and target new market segments.
Enhanced Customer Insights: Gain deeper understanding of consumer behavior.
Risk Mitigation: Diversify customer base to withstand market fluctuations.
By effectively segmenting users, businesses can enhance their marketing efforts, product offerings, and overall customer satisfaction.

6. Conclusion

This analysis successfully segmented users based on their interests in various types of POIs using clustering techniques. The exploratory data analysis and correlation matrix provided valuable insights into the dataset, while K-Means clustering emerged as the recommended method for user segmentation due to its higher Silhouette score.

The identified user segments can be leveraged by businesses to tailor their marketing strategies and service offerings, ensuring a more personalized and satisfying experience for their customers. By understanding the diverse needs and preferences of different user groups, businesses can optimize their resource allocation, product development, and marketing efforts.

Future research could explore the integration of additional user data, such as demographic information or behavioral patterns, to further refine the user segmentation process. Additionally, the application of advanced clustering techniques or methods could potentially improve the clustering performance and provide more robust and interpretable results.

7. References

1. RMIT Study material (COSC2670)
2. https://builtin.com/data-science/elbow-method
3. https://archive.ics.uci.edu/dataset/476/buddymove+data+set
4. https://rpubs.com/jabirkangarli/clustering-in-R